6.2 Radicals and Rational Exponents

Name _____ Period ____ Date _____

Algebra I 6.2 (30-minute plan) · Student sheet

I can…

- find nth roots.
- rewrite expressions between radical form and rational exponent form.
- evaluate expressions with rational exponents.

Vocabulary · fill in the missing word

nth root — if $b^n = a$, then b is an nth root of a, _____ $\sqrt[n]{a}$

Index — _____

Rational exponent — $a^{\left(\frac{1}{n}\right)}$ = _____ = $\left(\sqrt[n]{a}\right)^m$

Part 1 · Notes and examples (fill in each note, then work the example)

nth root: $\sqrt[n]{a}$ is a number whose _____ power is a. The n is called the _____.

Rational exponents: $a^{(1/n)}$ = _____ $a^{(m/n)}$ = _____

Example 1

Rewrite each expression in radical form.

a. $10^{\left(\frac{1}{2}\right)}$ b. $11^{\left(\frac{1}{5}\right)}$

Real nth roots: If n is odd, a has _____ real nth root. If n is even and a > 0, a has _____ real nth roots.

Example 2

Find the indicated real nth root(s) of a.

a. n = 3, a = −27 b. n = 4, a = 16

Reading a^(m/n): In $a^{(m/n)}$, the denominator tells the _____ and the numerator tells the _____.

Example 3

Evaluate each expression.

a. $16^{\left(\frac{3}{4}\right)}$ b. $27^{\left(\frac{2}{3}\right)}$ c. $32^{\left(-\frac{2}{5}\right)}$

Part 2 · Partner practice

Partner practice

Evaluate $64^{\left(\frac{2}{3}\right)}$.